

WASP-43b

R-1322

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_160229 · created 2026-07-31T19:41:12+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457447.8523 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.073 +/- 0.041
Transit depth (Rp/Rs)^2	0.53 +/- 0.6 [%]
Orbital Inclination (inc)	77.49 +/- 3.05
Airmass coefficient 1 (a1)	0.984 +/- 0.016
Airmass coefficient 2 (a2)	0.0108 +/- 0.0044
Scatter in the residuals of the lightcurve fit is	1.62 %
Best Comparison Star	#3 - [585, 336]
Optimal Aperture	4.88
Optimal Annulus	7.06
Transit Duration (day)	0.016 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.073 ± 0.041 vs 0.1594 (deviation 1.76σ)
Duration fit vs published	0.016 d vs 0.0512 d (deviation 1.64σ)
Mid-time O–C	-2.5 min
Depth SNR	1.5
Residual scatter	1.62 %

Light curve

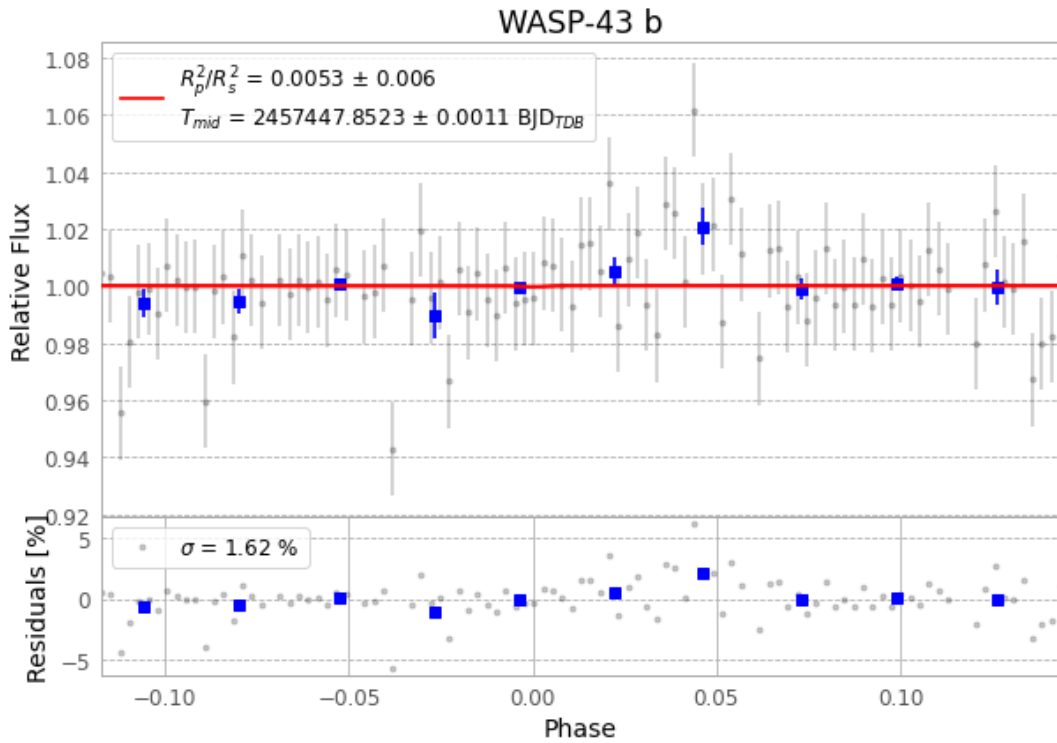


Figure 1 — FinalLightCurve_WASP-43 b_2016-02-28.png (SHA-256 f64f5645345f80a0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [444, 327], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 37e534205c0e38cc...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

103 FITS frames (68 MB), WASP-43160229060012.FITS → WASP-43160229110612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-160229.log (6154add6f76a...), FinalLightCurve_WASP-43 b_2016-02-28.png (f64f5645345f...), FinalLightCurve_WASP-43 b_2016-02-28.csv (7f2ba2d9b879...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 19:41Z.